

- Create a Git Repository (private) in GitHub.
- Download the origin repository to your machine using: *git clone <repo-name>*
- Now in your cloned local repository,
 - Design your assembler i.e., the file *assembler.sh* as per the specifications and pointers provided.
 - Throughout the design process, use the following commands when appropriate.

git add assembler.sh

git commit -m "any appropriate message"

git push

- Rubrics: **(2 points - bonus points)**

your code will be assessed against the following test cases.

1. **No argument (1 point):** the following command

bash assembler.sh

should return exit code 1 (error condition) and

display the message “usage: no argument is provided” on STDOUT.

2. **More than one argument (1 point):** the following command

bash assembler.sh a.vsc b.vsc

should return exit code 1 (error condition) and

display the message “usage: more than one arguments are provided” on STDOUT.

3. **Input is not a file or it does not exist (1 point):** the following command

bash assembler.sh ~

should return exit code 1 (error condition) and

display the message “usage: input is not a file or it does not exist” on STDOUT.

4. **Invalid file extension (1 point):** the following command

bash assembler.sh a.vs

should return exit code 1 (error condition) and

display the message “usage: input does not have the extension .vsc” on STDOUT.

5. **Empty file (1 point):** the following command (the file empty.vsc is attached)

bash assembler.sh empty.vsc

should return exit code 1 (error condition and

display the message “usage: the file is empty – no .bin file is produced” on
STDOUT.

6. **QUIT program (2 points):** the following command (the file quit.vsc is attached)

bash assembler.sh quit.vsc

should return exit code 0 (success)

and display the messages on STDOUT

“It is a QUIT program”

“The content of the .bin file is”

20

00

7. **ADD/SUB program (3 points):** the following command (the file add.vsc is attached)

bash assembler.sh add.vsc

should return exit code 0 (success)

and display the messages on STDOUT

“It is an ADD/SUB program”

“The content of the .bin file is”

7f

0c

04

00

0c

01

08

0e

24

00

20

00

- After your work is done, before the deadline 20-09-2026, submit your code *assembler.sh* in a zipped folder under Assignment 1 in **Canvas**. Along with this, add your marker to your repository on GitHub.

[Inviting collaborators to a personal repository - GitHub Docs](#)

- And in the same week, meet with your tutor/marker for a code review and discussion session (**it is compulsory part** so must be attended, no score can be awarded without this. At most – **3 points**)

Assignment 1 allows simple extension – informal arrangement between the UC (myself) and the student (yourself). It can be up to 2 days. You do not need to email me. It would be automatically applied to everyone. If you have a disability academic plan approved, the extension of 7 days is automatically approved.